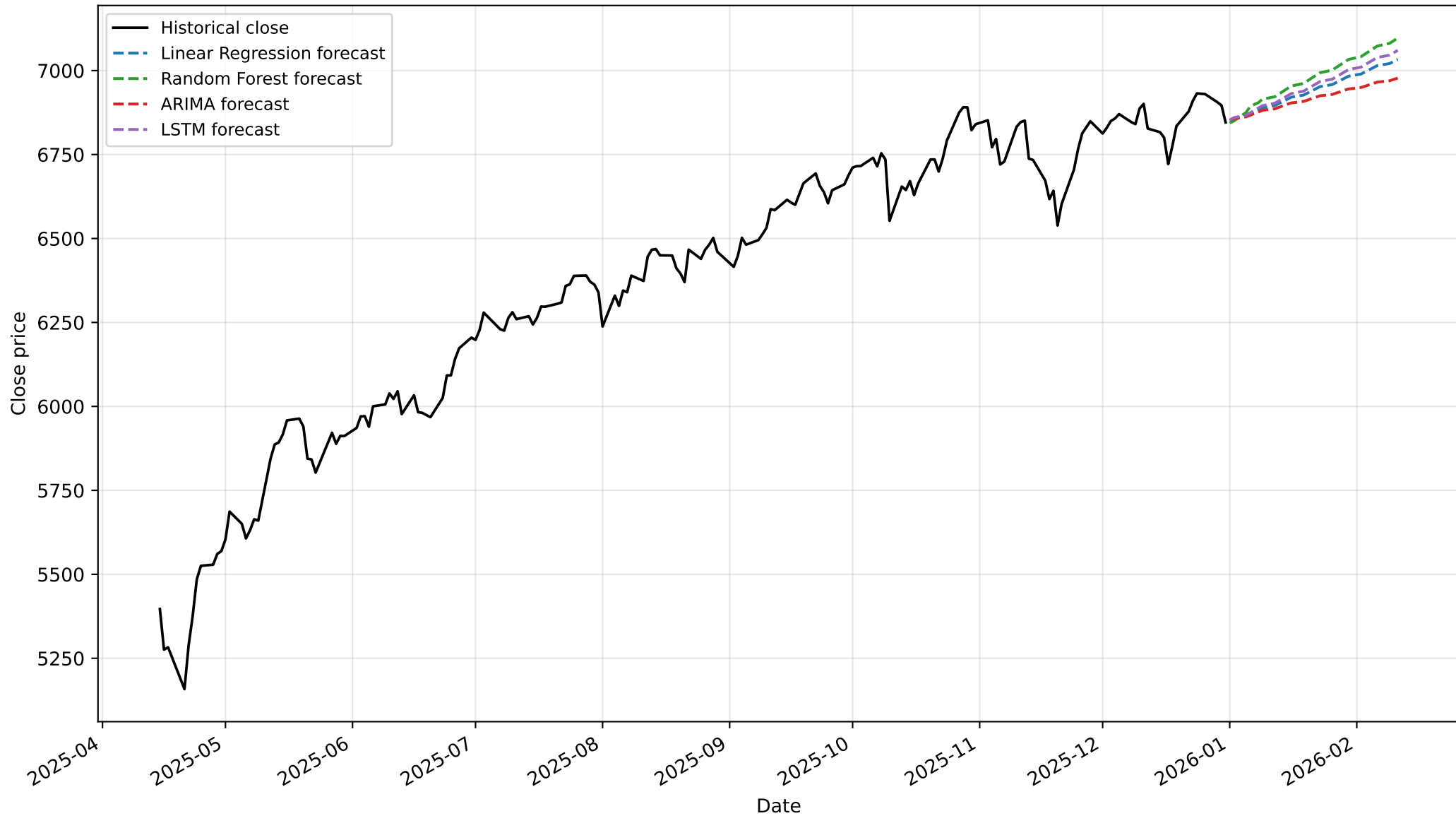


**PDA3 Batch Analysis Report**  
**Symbol: ^GSPC**  
**Source file: ^GSPC260101.csv**  
**Generated: 2026-09-26 05:00 UTC**  
**Observations used: 752**

Latest indicators  
Last Close: 6845.5  
Change 30D: 2.59  
Sma 20: 6856.23  
Sma 50: 6802.55  
Volatility: 15.07

*Estimates are educational model outputs, not investment advice. Historical accuracy does not guarantee future performance.*

^GSPC — Historical Close and Model Forecasts



## Model validation metrics (walk-forward holdout)

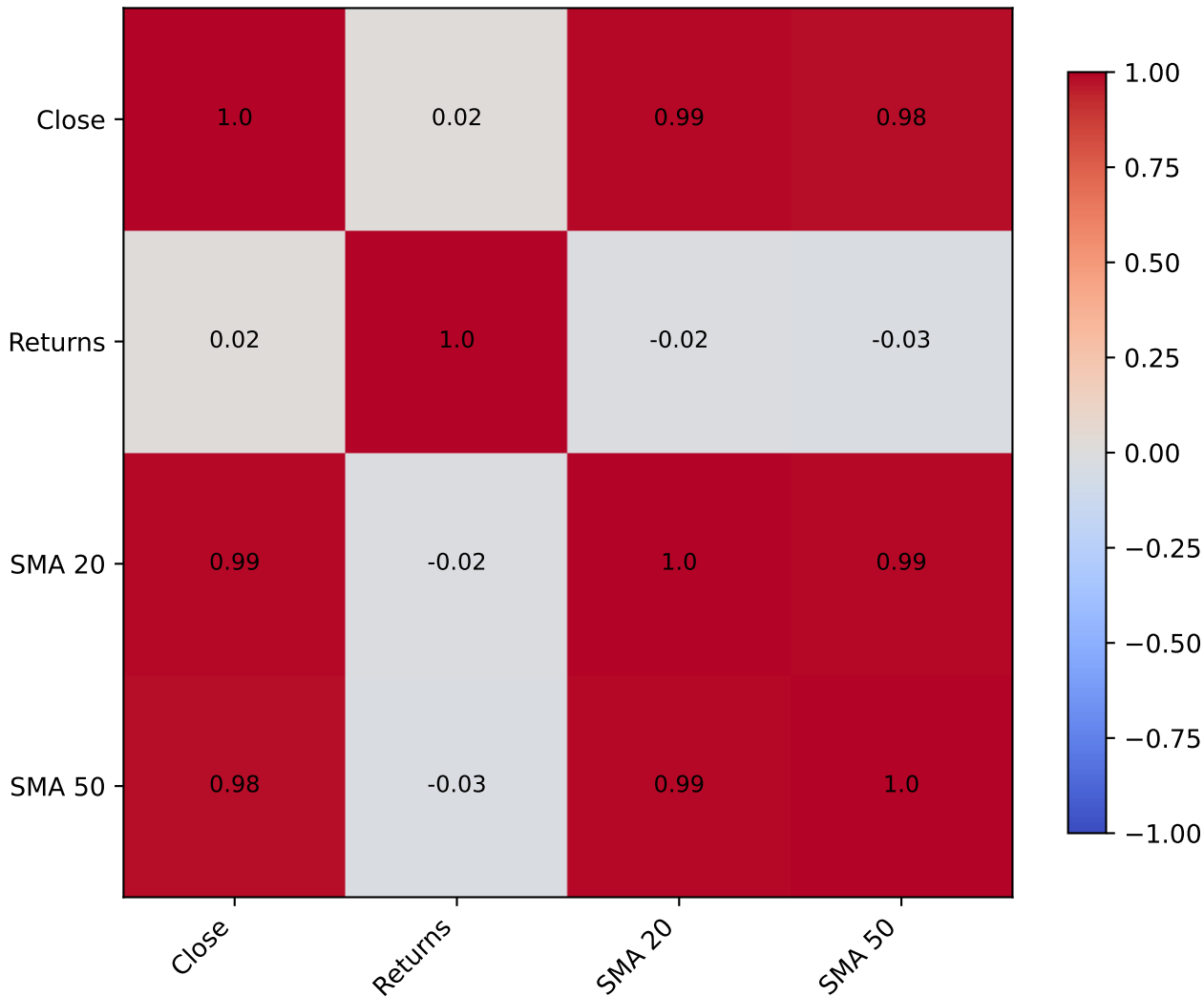
| Model                | MAE           | RMSE          | MAPE (%)    | R <sup>2</sup> | Notes |
|----------------------|---------------|---------------|-------------|----------------|-------|
| Linear Regression    | 59.551        | 70.672        | 0.88        | 0.4875         |       |
| <b>Random Forest</b> | <b>49.508</b> | <b>63.666</b> | <b>0.73</b> | <b>0.5841</b>  |       |
| ARIMA                | 80.869        | 89.417        | 1.19        | 0.1796         |       |
| LSTM                 | 74.128        | 89.42         | 1.09        | 0.1796         |       |

*R<sup>2</sup> below 0 means a model performed worse than simply predicting the average price over the validation window — common when a stock reverses or plateaus after a strong trend.*

Forecast table (next trading days)

| Date       | Linear Regression | Random Forest | ARIMA   | LSTM    |
|------------|-------------------|---------------|---------|---------|
| 2026-01-01 | 6851.77           | 6844.02       | 6851.78 | 6852.66 |
| 2026-01-02 | 6858.03           | 6849.7        | 6856.62 | 6859.82 |
| 2026-01-05 | 6864.3            | 6875.25       | 6862.4  | 6866.97 |
| 2026-01-06 | 6870.56           | 6892.92       | 6866.93 | 6874.13 |
| 2026-01-07 | 6876.83           | 6899.01       | 6872.33 | 6881.29 |
| 2026-01-08 | 6883.1            | 6904.46       | 6876.65 | 6888.45 |
| 2026-01-09 | 6889.36           | 6915.04       | 6881.75 | 6895.61 |
| 2026-01-12 | 6895.63           | 6922.02       | 6885.92 | 6902.76 |
| 2026-01-13 | 6901.89           | 6930.14       | 6890.8  | 6909.92 |
| 2026-01-14 | 6908.16           | 6938.01       | 6894.87 | 6917.08 |
| 2026-01-15 | 6914.43           | 6945.97       | 6899.58 | 6924.24 |
| 2026-01-16 | 6920.69           | 6953.92       | 6903.58 | 6931.4  |
| 2026-01-19 | 6926.96           | 6961.65       | 6908.16 | 6938.55 |
| 2026-01-20 | 6933.22           | 6969.55       | 6912.11 | 6945.71 |
| 2026-01-21 | 6939.49           | 6977.74       | 6916.59 | 6952.87 |
| 2026-01-22 | 6945.76           | 6985.51       | 6920.52 | 6960.03 |
| 2026-01-23 | 6952.02           | 6993.41       | 6924.91 | 6967.19 |
| 2026-01-26 | 6958.29           | 7001.42       | 6928.82 | 6974.35 |
| 2026-01-27 | 6964.55           | 7009.37       | 6933.15 | 6981.5  |
| 2026-01-28 | 6970.82           | 7017.33       | 6937.05 | 6988.66 |
| 2026-01-29 | 6977.08           | 7025.28       | 6941.33 | 6995.82 |
| 2026-01-30 | 6983.35           | 7033.24       | 6945.23 | 7002.98 |
| 2026-02-02 | 6989.62           | 7041.19       | 6949.47 | 7010.14 |
| 2026-02-03 | 6995.88           | 7049.15       | 6953.38 | 7017.29 |
| 2026-02-04 | 7002.15           | 7057.1        | 6957.58 | 7024.45 |
| 2026-02-05 | 7008.41           | 7065.06       | 6961.49 | 7031.61 |
| 2026-02-06 | 7014.68           | 7073.01       | 6965.66 | 7038.77 |
| 2026-02-09 | 7020.95           | 7080.97       | 6969.58 | 7045.93 |
| 2026-02-10 | 7027.21           | 7088.92       | 6973.73 | 7053.08 |
| 2026-02-11 | 7033.48           | 7096.88       | 6977.65 | 7060.24 |

Correlation matrix

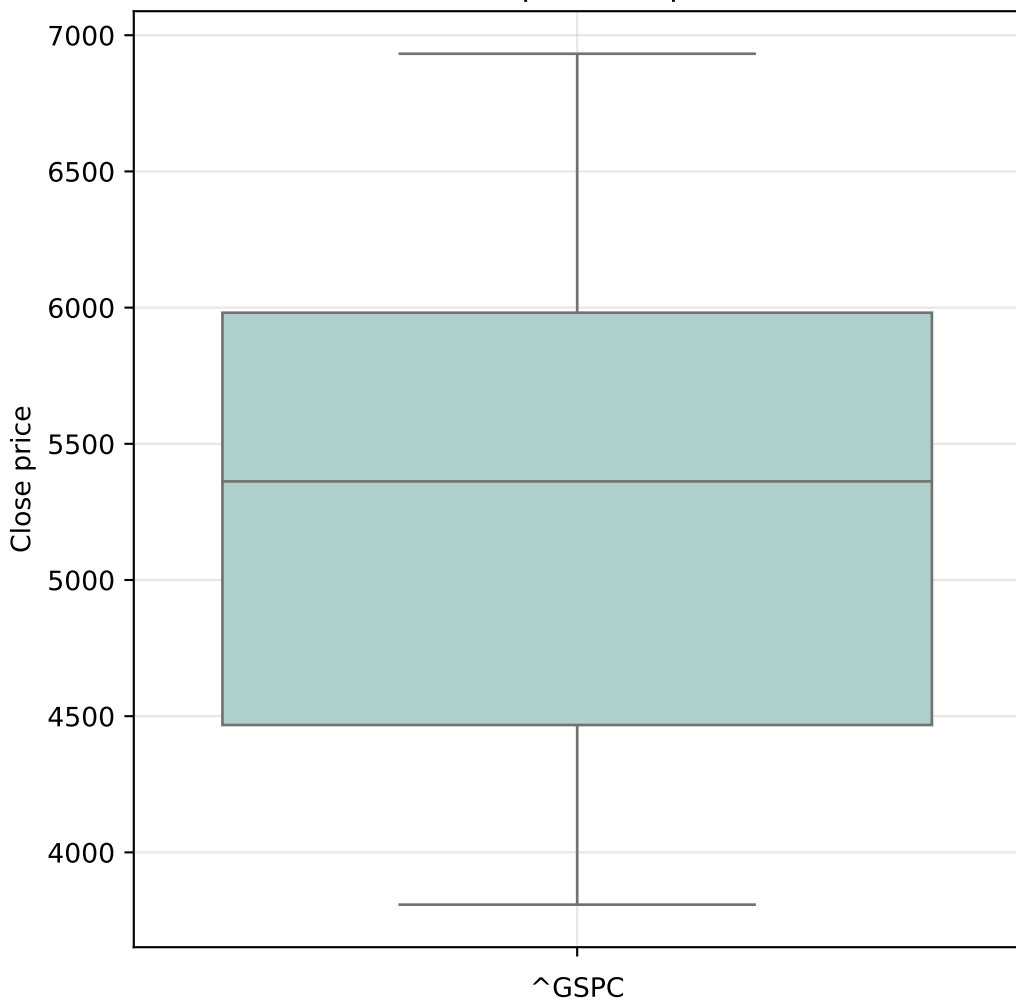


## ^GSPC — Source data statistics (close price)

| Statistic               | Value    |
|-------------------------|----------|
| Count                   | 752      |
| Mean                    | 5309.926 |
| Median                  | 5362.075 |
| Std Dev                 | 873.397  |
| Min                     | 3808.1   |
| Max                     | 6932.05  |
| Range                   | 3123.95  |
| Q1 (25%)                | 4467.642 |
| Q3 (75%)                | 5981.333 |
| IQR                     | 1513.69  |
| Skewness                | 0.053    |
| Kurtosis                | -1.155   |
| Coeff. of Variation (%) | 16.448   |

# $\hat{GSPC}$ — Source data range and distribution

## Close price boxplot



## Close price distribution (violin plot)

